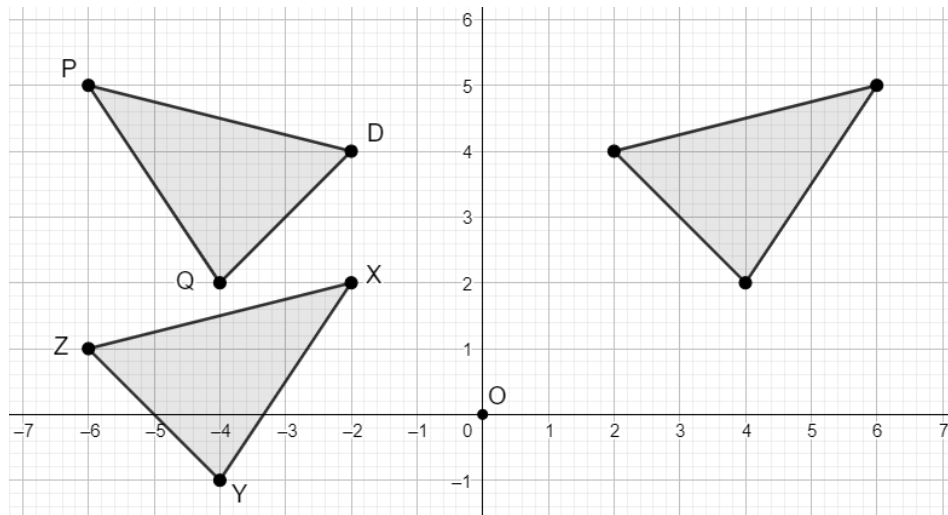


Geometry
Unit 6
Lesson 4 Practice

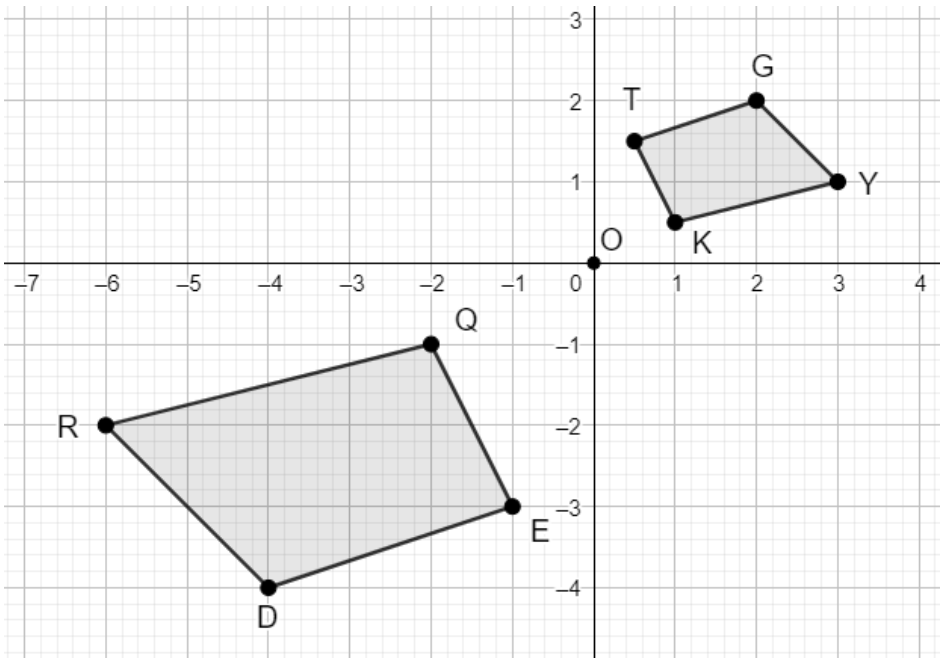
Name _____
Date _____
Period _____

$\triangle XYZ$ is the result of a sequence of transformations of $\triangle PDQ$. The unlabelled triangle is the first transformation.



1. Describe the first transformation of $\triangle PDQ$.
2. Write the transformation algebraically.
3. Describe the second transformation that results in $\triangle XYZ$.
4. Write the second transformation algebraically.
5. Explain if the sequence of transformations preserves distance.

KTGY is the result of a sequence of transformations of QEDR.



Ian and Natalie were determining the sequence of transformations, but each came up with a different sequence. Complete the table for the missing portions of their transformations.

			First Transformation	Second Transformation
6.	Ian's Sequence	Written Description	Rotate 180° about the origin.	
7.		Algebraic Description		$(x,y) \rightarrow (\frac{1}{2}x,\frac{1}{2}y)$
8.	Natalie's Sequence	Written Description	Dilate using a scale factor of $\frac{1}{2}$ centered at the origin.	
9.		Algebraic Description		$(x,y) \rightarrow (-x,-y)$

10. Considering both proposed sequences, you can conclude that

- ☐ Ian's sequence is

☐ Natalie's sequence is

☐ both sequences are

correct.